A proof of Hardin's empirical column formulas for OEIS A188866, via ranges of lazy walks, formalized in Lean 4

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Abstract

Let $T(n,k)$ count the $n \times k$ binary arrays that avoid the pattern 01 vertically, diagonally and antidiagonally. In other words, a 0 in row i forces 0s in the three cells below it in row $i+1$. This is OEIS A188866. Its column $k=4$ is OEIS A188861. R. H. Hardin listed eight empirical column formulas ($T(n,4) = 27n-13$ for $n>1$, …, $T(n,8) = 2187n-3495$ for $n>5$), and they are still marked "Empirical" on OEIS. We prove one uniform closed form:

Theorem 1. *For all $k \geq 1$ and $n \geq k-2$,*

$$T(n,k) = 3^{k-1}(n+1) - R_{k-1}, \qquad R_L = \sum_{s\in\{-1,0,1\}^L} \big(\max_i S_i - \min_i S_i\big),$$

where $S_0 = 0$, $S_i = s_1 + \cdots + s_i$. So R_L is the total range of all 3^L lazy walks of length L.

$R_0, \ldots, R_9 = 0, 2, 10, 40, 146, 508, 1716, 5682, 18546, 59884$. Plugging in $R_0, \ldots, R_7$ gives all eight of Hardin's column formulas, each with exactly Hardin's range of validity. The whole development is machine-checked in Lean 4 + Mathlib with no `sorry` and no axioms beyond Lean's standard three.

1 Statement and context

OEIS A188866 (R. H. Hardin, 2011): $T(n,k)$ as above. It lists the formulas

$$\begin{aligned}
&T(n,1) = n+1, && T(n,2) = 3n+1, && T(n,3) = 9n-1,\\
&T(n,4) = 27n-13 \ (n>1), && T(n,5) = 81n-65 \ (n>2), &&\\
&T(n,6) = 243n-265 \ (n>3), && T(n,7) = 729n-987 \ (n>4), &&\\
&T(n,8) = 2187n-3495 \ (n>5), &&&&
\end{aligned}$$

all labelled Empirical.

OEIS A188861: column 4, "Empirical: $a(n) = 27n-13$ for $n>1$". It also carries conjectured g.f. and recurrence from C. Barker (2018).

2 Proof

The predicate `Valid n k` in the formalization is the direct transcription of the OEIS definition. A 0 (false) at (i,j) forces 0 at $(i+1,j-1)$, $(i+1,j)$, $(i+1,j+1)$ whenever those cells exist. Rows

are indexed from 0 (top) to $n-1$ (bottom), and columns from 0 to $k-1$. This is equivalent to each column being an initial segment of 1s followed by 0s, with adjacent column heights differing by at most 1.

Step 1 (arrays $\leftrightarrow$ smooth words). Taking $j' = j$ shows each column is a block of 1s followed by 0s. Let $t_j \in \{0, \ldots, n\}$ be the number of 1s in column j. The diagonal conditions hold exactly when $|t_j - t_{j+1}| \leq 1$ for every j. If $t_j \geq t_{j+2}$, the 0 at row t_j of column j sits above a 1 in column j'. Conversely, if all adjacent letters differ by at most 1, no such configuration can occur. So $T(n,k)$ counts words $t \in \{0, \ldots, n\}^k$ with adjacent letters differing by at most 1. This bijection was noted on OEIS by A. Howroyd (2017). Such words are the "smooth words" of Knopfmacher, Mansour, Munagi and Prodinger.

Step 2 (fibres over step sequences). Write $L = k-1$ and $M = n$. A smooth word is fixed by its step sequence $s \in \{-1, 0, 1\}^L$ together with its first letter x, because $t_i = x + S_i$. The word stays inside $[0, M]$ exactly when $x \in [-\min S, M - \max S]$. That interval has $M + 1 - (\max S - \min S)$ points whenever $\max S - \min S \leq L \leq M+1$. Summing over the 3^L step sequences gives $T(n,k) = 3^L(M+1) - R_L$.

Step 3. Evaluating $R_0, \ldots, R_7$ (done in Lean by kernel computation) gives the eight formulas.

3 Lean formalization

Project: `NovelProof/` (Lean v4.35.0-rc4, Mathlib v4.35.0-rc4, exact revisions locked in `lake-manifest.json`). Source: https://github.com/het4rk/OEIS-A188866-proof.

- `NovelProof/Words.lean`: smooth words `WValid M L` and their count `W M L`; step sequences `D L` $= \{-1,0,1\}^L$; partial sums `P`; `hi`, `lo` (max and minus min of the walk); total range `R L`. The fibre count `fiber_card` (each step sequence d has exactly $M + 1 - (\texttt{hi}\, d + \texttt{lo}\, d)$ starting points, using `hi_add_lo_le : hi d + lo d` $\leq L \leq M+1$, so the interval is never empty) and the closed form `W_formula : L ≤ M+1 → W M L = 3^L*(M+1) - R L`.
- `NovelProof/Arrays.lean`: the array predicate `Valid n k` and `T n k`, the bijection `T_eq_W : T n (L+1) = W n L` (via `toArr`/`fromArr`), and `T_formula : L ≤ n+1 → T n (L+1) = 3^L*(n+1) - R L`.
- `NovelProof/Corollaries.lean`: `R_values` ($R_0, \ldots, R_7$ by kernel computation) and Hardin's eight column formulas `T_col1`, …, `T_col8`, each with Hardin's range of validity.

`lake build` completes with no errors. `lake env lean Axioms.lean` reports only `[propext, Classical.choice, Quot.sound]` for `T_eq_W`, `T_formula`, `T_col4` and `T_col8`. There are no `sorry`s. CI (GitHub Actions) rebuilds and re-checks axioms on every push.

4 Novelty search (honest assessment)

What I checked: the OEIS entries A188866, A188861 and A188860 and their cross-references, OEIS searches for the sequences R_L and $c_k = 3^{k-1} - R_{k-1}$ (no hits for 0,2,10,40,146,508,1716 or 1,1,-1,-13,-65,-265,-987), and web searches. I also checked the smooth-words literature: A. Knopfmacher, T. Mansour, A. Munagi, H. Prodinger, Smooth words and Chebyshev polynomials, arXiv:0809.0551.

Known: the bijection to smooth words (Howroyd on OEIS), and a full generating function and trigonometric formula for smooth words (Knopfmacher et al. 2008). Each fixed-k column formula

can therefore be derived from known results in principle, so the column formulas are not deep open problems. Even so, OEIS still lists them as unproven ("Empirical").

Believed new: (i) the uniform statement $T(n,k) = 3^{k-1}(n+1) - R_{k-1}$ for $n \geq k-2$, which identifies the constant term as the total range of lazy walks; (ii) the integer sequences R_L and c_k, which are absent from OEIS; (iii) the first formal verification of these formulas, as far as I can find.

Confidence: high that the theorem is correct (machine-checked). The novelty is modest: this proves a family of OEIS conjectures and adds a new closed form. It is not a breakthrough. The constant could plausibly be extracted from the asymptotic analysis in Knopfmacher et al., though I did not find it stated there in this form.

5 Limitations / further work

No closed form or generating function for R_L is proved. Hardin's row recurrences (fixed n, varying k) are not formalized. They follow from the transfer matrix $I + A(P_{n+1})$ and the Chebyshev g.f., but are not formalized here.

Possible next steps: submit the proof and the sequences R_L, c_k to OEIS, and formalize the row recurrences.